



Bitcoin's "ultimate experiment in capitalism," explained

"It's all speculation at this point."

By Emily Stewart | Jan 3, 2018, 8:30am EST



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In 2013, a friend and I were convinced by a tech-savvy acquaintance to invest in something called Litecoin — basically, Bitcoin's less expensive little sister. We put \$100 each into a random exchange, which at the time got us about 60 litecoins total, give or take, and forgot about them.

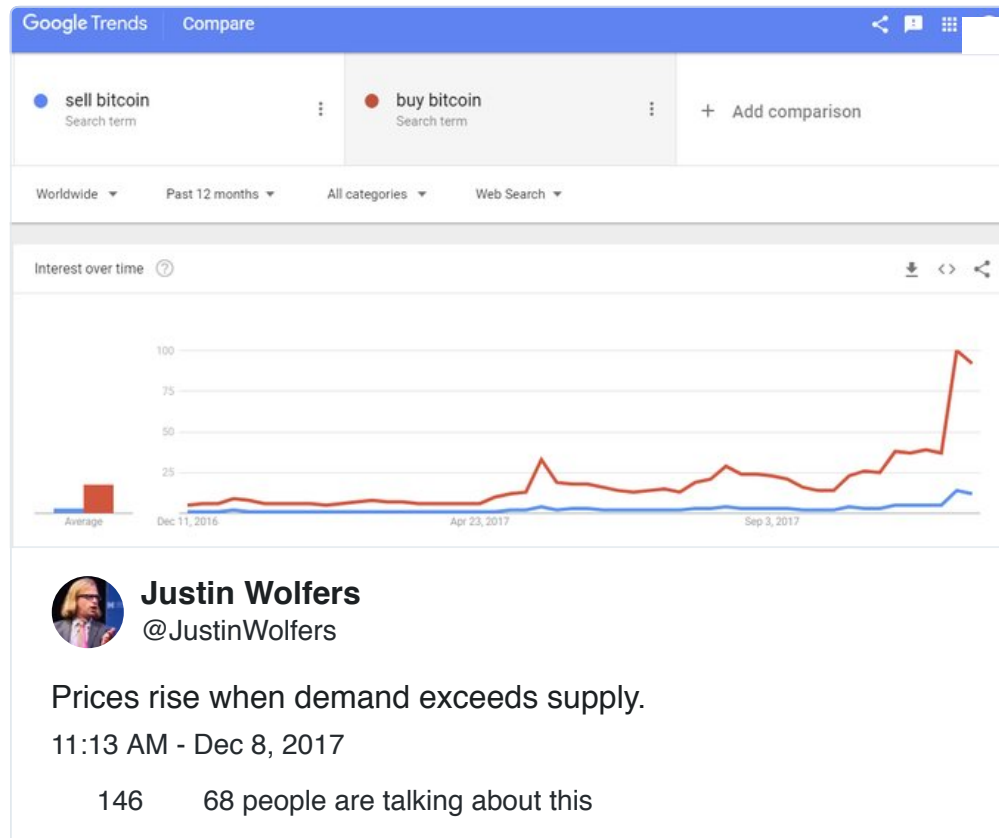
Over the years we'd talked about getting them out, but we never did. And then over the summer the federal government shut down the exchange over its ties to money laundering.

As Bitcoin's price went on a wild roller coaster ride in recent months, we joked about getting back in. But our cryptocurrency speculation days are probably over — we're not as eager to pay thousands of dollars for internet money that no one except for a select few seems to understand.

The thing about Bitcoin — the thing about currency, broadly — is that its value depends entirely on what people are willing to pay for it. And right now, people are willing to pay a lot

for Bitcoin, with varying degrees of enthusiasm. Its price climbed more than 1,000 percent in 2017, starting the year at about \$1,000 and in mid-December nearly grazing \$20,000. It ended the year around \$13,000.

Google searches for Bitcoin are, for the first time, surpassing searches for Donald Trump. And as University of Michigan economist Justin Wolfers **noted on Twitter**, people are searching for how to buy Bitcoin, not sell it. Prices rise when demand exceeds supply, and more people wanting to buy Bitcoin explains its meteoric price rise.



But there's no clear explanation for *why* there's this sudden spike in interest, and even many Bitcoin true believers are wary of what's going on. Bitcoin is notoriously volatile and has seen multiple booms and crashes. In April 2013, it lost **more than half** of its value overnight. In 2014 when one of its major exchanges **lost \$400 million**, Bitcoin's price fell by half again.

And the volatility hasn't settled. Last summer, Bitcoin **dropped by 40 percent** on concerns over a clampdown in China. It had **multiple** price corrections of more than 25 percent in 2017. For some perspective, Black Tuesday, the Wall Street crash that precipitated the Great Depression, saw stocks fall 12 percent in a single day.

“It’s all speculation at this point,” said Mark Zandi, chief economist at Moody’s Analytics. **“Beanie Baby, Pets.com, tulip bubble** speculation.”

Bitcoin’s volatility is, in part, what has drawn investors, speculators, and, increasingly, regulators. The price of Bitcoin surpassed \$10,000 for the first time in late November and has neared, though not surpassed, \$20,000. But the ride hasn’t been a smooth one. For example, it topped \$17,000 for the first time on December 8 and by December 10 was back in the \$13,000 range. On December 19, its price fell by 10 percent in a **matter of minutes** when a popular US-based exchange began trading bitcoin cash, a spinoff of the original Bitcoin.

“It’s insane,” said Brayton Williams, the co-founder of Boost VC, a California-based accelerator that invests heavily in Bitcoin and blockchain, the ledger technology upon which Bitcoin is based. “For us, we have this weird dilemma where we’ve been fighting this battle for the last four and a half years, and now it’s like, did we win?”

Bitcoin appears to be in **bubble territory** — not only because of its price runup but also given the speculation, volatility, and new players in the space.

But the wild ride might be worth it. Bitcoin believers say it could be the future of money — but even if it isn’t, it’s quite a world all on its own. Bitcoin has hit the mainstream in recent months after garnering more attention in the media, on Main Street, and on Wall Street, and has taught investors a lesson in its potential as well as its pitfalls. It’s very possibly in bubble territory, but that might not diminish its eventual impact on finance and technology.

So what is Bitcoin, anyway?



Alain Pitton, NurPhoto/Getty Images

Bitcoin's value fluctuates so rapidly because it's a currency that's not backed by a government or a physical measure of worth — it's what people call a cryptocurrency, a digital or virtual currency that uses cryptography (writing or solving codes) for security.

It's completely decentralized, meaning no one government or entity is in charge of it, and it is built off a network called a blockchain, a public ledger of peer-to-peer transactions. You can think of it sort of like an **endless Excel spreadsheet** everyone shares. Because transactions on the blockchain are publicly verified, market participants can keep track of transactions without central record keeping. A block that's been altered would be obviously out of place and therefore easy to spot.

Bitcoin has been around for less than a decade. The concept first showed up online in a **2009 white paper** by a person using the alias Satoshi Nakamoto whose identity has never been revealed. The paper outlines an idea for a version of electronic cash that would allow online payments to be made from one person to another without a financial institution or some other third-party arbiter in the middle.

Bitcoin is created through a process called mining, which Vox's **Umair Irfan** recently described as “finding solutions to complicated math problems that become progressively more difficult.” Coins are awarded to computers that verify transactions with an algorithm that gets more complex over time.

When Bitcoin started out, coins could be mined on home computers and were awarded at about **\$2** each. Now mining is a much more complicated — and energy-intensive — process involving giant **mining farms** of thousands of computers across the globe. According to **one recent report**, Bitcoin mining is now consuming more electricity than 159 countries, raising concerns about just how much energy it's taking up.

When Bitcoin first emerged, it was largely reserved to the **dark corners of the internet** and, often, the **black market**. Because payments with it are essentially impossible to trace, a lot of its use case has been for buying and selling drugs and weapons and carrying out other illegal transactions. The now-defunct online black market **Silk Road**, which the government shut down in 2013, ran its transactions on Bitcoin.

Beyond the United States, Bitcoin is also popular in **China, South Korea, Japan**, Russia, Nigeria, South Africa, and Venezuela for a variety of reasons. In Venezuela, for example, hyperinflation and harsh currency controls from the government has made Bitcoin a more stable investment and a way to get money in and out of the country. In **Japan**, Bitcoin has been recognized as legal tender, and financial regulators have officially recognized multiple cryptocurrency exchange operators.

Some see the underlying blockchain technology as Bitcoin's true value, envisioning it as a way to disrupt **how international trade deals are financed and negotiated** and **the way health care providers share information**.

Bitcoin true believers see Bitcoin as having the potential for much more. They hope the current craze will fuel the system they eventually want to create — an “open consensus mechanism” allowing people all over the world to reach a trustworthy agreement without there being a central planner or authority, said Peter Van Valkenburgh, research director at the public policy advocacy group Coin Center.

Some even subscribe to a theory of “hyperbitcoinization,” where Bitcoin essentially takes over the global financial system. In such a scenario, Bitcoin would become so valuable that it would upend other currencies throughout the world, and essentially everything would

run on Bitcoin. “As Bitcoin grows, its incentive for you to adopt it gets even greater,” said Steven McKie, founding partner at the crypto firm Amentum.

So while the current Bitcoin craze is to some observers a textbook example of speculation gone wild, there are also deeper implications. For true believers, it's not just about the price of a single investment, but instead a system that they believe could circumvent traditional governments by creating a decentralized, global financial system that no single entity oversees. For them, there's a lot more at stake than the desire to make a quick buck on a current trend.

Bitcoin's mainstream takeoff is a combination of price, investor interest, and a fear of missing out

The safest way to trade bitcoins is on cryptocurrency exchanges, which are sort of like the New York Stock Exchange or Nasdaq, where stocks are bought and sold, but for digital currency. And the exchanges have seen an enormous influx of interest and money in recent months.

Right now, Bitcoin feels a little like the gold rush, or like the **Dutch tulip mania**, with fortunes being made and lost overnight.

Coinbase, one of the biggest cryptocurrency exchanges in the US, added **100,000 new accounts** in a single day in November and became the **most downloaded app** in the Apple app store in December. The exchange has struggled to keep up — its platform has **gone down** on multiple occasions, overwhelmed by the influx of new users flocking to it to make a quick buck. A Coinbase representative said the company's executives were too busy for an interview for this story.

“What you're seeing is a combination of a lot more visibility, which is giving it a lot more validation to investors and to the public,” said Jack Tatar by way of explanation. Tatar is an angel investor and the author of ***Cryptoassets: The Innovative Investor's Guide to Bitcoin and Beyond***, a book on Bitcoin investing. “And then as the price runs, it's like a domino effect,” he said.

As the price goes up, and people see more and more investors jumping in, they hop in too. Nick Colas, co-founder of DataTrek Research, a market insight and research firm in New York, told me that people appear to be buying Bitcoin with their credit cards — a risky proposition.

People are afraid of missing out on a new trend that might make them piles of money, said Rob Long, a partner at Bell Nunnally & Martin and former attorney at the Securities and Exchange Commission, the Department of Justice, and the Financial Industry Regulatory Authority. “It’s supply and demand — limited supply, and as this demand gets elevated, it becomes more newsworthy,” he said.

There is probably some **“greater fool theory”** in play, the idea that people invest in something not because of its value but instead because they assume they’ll later be able to sell it to someone — the greater fool — willing to pay a higher price.

“The chart of Bitcoin is parabolic, and nothing that goes parabolic maintains that trajectory,” said Sam Stovall, chief investment strategist at the investment research firm CFRA Research.

Bitcoin markets are developing — but banks and regulators are pretty wary of them



Cboe Global Markets launched a futures market on Bitcoin in late 2017. | Scott Olson/Getty Images

Right now, the primary way of investing in Bitcoin is buying and selling bitcoins themselves. But more traditional futures markets are starting to get involved — opening the door to more traditional investors, such as hedge funds and institutions.

Two major exchanges — CME Group and Cboe Global Markets — have launched futures markets on Bitcoin.

Futures markets are essentially an agreement to buy or sell something at a future date at an agreed-upon price and exist for a wide range of markets. Cboe, one of the companies offering Bitcoin futures trading, also has futures trading on stock market volatility, for example. They provide a way for traders not to invest directly in Bitcoin itself but instead wager on where they think it's going, whether they believe the price will go up or down.

The futures markets are expected to draw more institutional investors into the Bitcoin space because it **lets them hedge their exposures** to protect themselves against Bitcoin's wild price swings. They can buy Bitcoin (or Bitcoin futures contracts wagering that the price will go up) and couple that with a short contract on Bitcoin, or a bet that the price will go down. (Remember **The Big Short**, where a bunch of guys made a bunch of money betting the housing market would collapse? Like that.)

Before the futures markets, the main way of shorting Bitcoin was hackers infiltrating exchanges to drive down prices and buy when prices are artificially low. Proponents say these new markets will reduce volatility in the underlying market and sort of tame Bitcoin, but whether that's actually how that will work is unclear. **Trading volumes** have been sluggish, meaning investors are still pretty reluctant.

And if you're investing in Bitcoin, when it comes to regulation, you're mostly on your own.

The Commodity Futures Trading Commission, the federal regulatory body that oversees the Bitcoin futures markets, has **taken steps** to reduce the risk of market manipulation and launched a **website** to educate traders. But even the CFTC's chair, Christopher Giancarlo, has warned investors that bitcoin markets and exchanges "remain largely unregulated markets" over which his agency has "limited statutory authority."

Regulators are still playing catch-up. "They're definitely ramping up their activity in this kind of space, but it takes a little while for the machines to get fired up," Long said.

JPMorgan Chase, Bank of America, Citigroup, and Royal Bank of Canada told customers they wouldn't offer them access to the first bitcoin futures markets when they went live,

the Wall Street Journal has reported. Goldman Sachs said it would offer limited access for certain customers, but according to **Bloomberg**, the bank has demanded that some of its clients set aside funds equal to the full value of their bitcoin futures trades as a condition for doing the transaction, meaning Goldman is still very nervous about the potential for big losses in Bitcoin futures.

Interactive Brokers Group, an online brokerage, is **warning** customers in a disclosure form that trading bitcoin futures is “especially risky” and “there may be no fundamental or economic basis for valuation of [b]itcoins and their prices may move randomly.”

Bitcoin advocates believe the futures markets could usher in Bitcoin exchange-traded funds (ETFs), investment funds traded like stocks on stock exchanges. So far, though, regulators have been wary of those too.

The Winklevoss twins — made famous for claiming Mark Zuckerberg **stole their idea for Facebook** in a lawsuit, in which they **reached a settlement** for \$65 million in cash and Facebook stock — have been unsuccessfully pushing for regulators to approve their proposed Bitcoin ETF since 2013. The Securities and Exchange Commission, which oversees ETFs, **rejected** their bid in March, saying that “significant markets for bitcoin are unregulated.”

“If there’s ever been an ultimate experiment in capitalism, this is it,” Tatar, the Bitcoin investment book author, said.

That means you invest at your own risk, including and perhaps especially when it comes to Bitcoin and other cryptocurrencies. The New York Stock Exchange has **“circuit breakers”** that halt trading when panic ensues. In the world of Bitcoin, there’s no such thing. If Bitcoin crashes tomorrow, and many people think it could, there’s nothing to stop investors from losing their shirts.

And if your coins are stolen, there’s not much you can do about it either — there is no FDIC insurance for cryptocurrency.

“Think about it the way you would think about a stack of \$100 bills in your sock drawer,” Colas said. “If it’s stolen, there’s no recourse.”

Mt. Gox, one of the biggest exchanges in the world, went down in 2014 after losing hundreds of thousands of bitcoins that were likely stolen. This year, US federal authorities **shut down** BTC-e, one of Bitcoin’s largest and oldest exchanges, for violations of anti-

money laundering laws. It has since launched a **new website** and has **promised** to return the funds users lost.

While reporting for this story, I discovered the Litecoin my friend and I thought we had lost still exists out there on the internet. We still can't figure out how to get it out.

As the Bitcoin arena has matured, so to speak, so have the exchanges. Platforms such as Coinbase comply with anti-money laundering and know-your-customer laws that require businesses to identify and verify the identities of their clients. Even so, exchanges can and have been hacked, and regulators could always, in theory, determine at any time that an exchange is unlawful. It's a high-risk game, with big money on the line.

Is Bitcoin a bubble? Probably — but broadly speaking, that might be okay.



Peter Thiel's Founders Fund has reportedly made a major investment in Bitcoin.

Bitcoin is still too small a segment of the global economy to represent a real threat to the financial system if it crashes — cryptocurrencies' total value is about \$500 billion, a small fraction of the roughly \$70 trillion global GDP. But taking out debt to buy an asset as volatile as Bitcoin — as some investors seem to be doing with their credit cards — is risky on a personal finance level.

“When you add leverage to any financial asset that’s risen quickly, you’re going to get unintended consequences when it comes back down to earth,” Colas said.

Bigger institutional fish, like JPMorgan Chase CEO Jamie Dimon, appear to agree. Dimon has called Bitcoin a “fraud” that “won’t end well.”

Billionaire hedge fund manager Seth Klarman **has said** Bitcoin is a “trading sardine.” Jim Chanos, who became famous for negative calls on Enron and Tyco, has called Bitcoin “a speculative mania” and compared it to the '90s Beanie Babies craze.

That’s not to say there isn’t money to be made. Peter Thiel, a billionaire venture capitalist who was one of Facebook’s first investors and is close to President Donald Trump, has **reportedly invested** millions of dollars in Bitcoin through his Founders Fund venture capital firm.

Many of the people I talked to for this story agreed that Bitcoin is likely in bubble territory, but they also said that might not matter.

Bitcoin has seen plenty of bursts and busts before, and it’s survived. Other industries have made it through busts as well — **the railroad, real estate, the internet.**

“This is not unusual; when you have new technologies, you have this period where people kind of lose their minds and throw lots of money at it,” Zandi, the Moody’s analyst, said. “The internet today is a result of the speculation and overinvestment that occurred in the 1990s and early 2000s.”

“This will burst, just like the dot-com bubble burst, but some interesting things will emerge from that,” Grayson Earle, the co-creator of **Bail Bloc**, a cryptocurrency app that helps poor people make bail, said.

Anyone who says they know what happens next in Bitcoin is lying

The initial promise of Bitcoin was a mixture of everyday utility — **Overstock.com**, for example, accepts bitcoins, and Rand Paul **took donations** in them for his presidential

campaign — and ideology, the belief that Bitcoin and blockchain technology can overhaul the global financial system.

But the Bitcoin and broader cryptocurrency space still have a lot of kinks to work out.

Coinbase's platform has **gone down** on multiple occasions recently, overwhelmed by the influx of new users flocking to the platform to make a quick buck. Coinbase CEO Brian Armstrong penned a **message to users** essentially telling people to chill out.

"To me that says that there are some concerns about how well these exchanges work," Tatar said, adding that it could have an effect on futures markets as well. "I would be surprised if there are not some real bumps in the road here when this first starts, because I'm not sure that the whole infrastructure can handle this."

Cryptocurrency transaction speeds are still **significantly slower** than, say, credit cards. Bitcoin can handle a handful of transactions per second, Visa can handle thousands, and developers still aren't sure how to scale.

More broadly, the launch of the futures markets and a growing public interest in Bitcoin is setting up a new dynamic between the traditional investment community — the Wall Street banks, hedge funds, mom-and-pop investors — and the true believers, the people who see Bitcoin as much more than a new fun thing to try out.

"As the system grows and the likelihood of it scaling grows, and as new speculators come on board with piles of cash, it compounds together and has created this exponential growth," said McKie, the crypto firm partner. "Wall Street thinks that they're smart, but they're not crypto smart."

While speculators may come and go from the space to make fast cash — what's happening now — there is a subset of individuals who will keep their bitcoins, essentially, forever. The crypto community describes that as to **"hodl,"** meaning they resist the temptation to sell and essentially hold on to their coins at all costs.

Greg Xethalis, a lawyer at Chapman and Cutler in New York with expertise in financial services and emerging technologies, speculated that the "hodl" culture — which combines ideological motivations and a long-term investment view on Bitcoin — creates unique Bitcoin purchasing expectations among a certain subset of investors who are less likely to engage in short-term profit taking than traditional investors.

“It is remarkable in that it’s really not an asset; it’s just kind of a faith-based investment,” said Jack Ablin, chief investment officer at BMO Private Bank in Chicago.

The current craze has been about price and market speculation, but there are other applications for Bitcoin and the blockchain that could be much more meaningful. A number of sectors, including **banking, finance**, and **health care**, could potentially implement elements of blockchain technology. There are other, more socially minded applications as well.

One such example is **Bail Bloc**, the aforementioned cryptocurrency app for bail. Launched in November, the application uses a computer’s unused power to mine a cryptocurrency called Monero. At the end of the month, it exchanges that for US dollars and donates to the Bronx Freedom Fund to post bail for low-income people detained in New York. The project has already raised about \$4,800.

Earle, one of the app’s co-creators, told me he sees the app as a pivot away from the libertarian politics of Bitcoin’s roots to a more socially inclined, progressive use case. “It’s important for other voices to get involved in the [crypto] space right now, because if it does become a new financial system, then we’re screwed. It’s benefiting this cast of characters; it’s a bunch of white dudes getting rich,” he said.

Industry speculation is making virtually everyone in the space at least a bit nervous. Multiple countries have issued **cryptocurrency warnings** and even **considered shutting down** exchanges.

What’s happening is causing quite a bit of self-reflection in the cryptocurrency community itself. Vitalik Buterin, the programmer who founded Ethereum, another cryptocurrency, in a series of **tweets** recently wondered whether the community had “earned” its newfound notoriety.

“How many unbanked people have we banked?” he asked. “How much value is stored in smart contracts that actually do anything interesting? How many Venezuelans have actually been protected by us from hyperinflation?”